STATISTICAL MODEL OF NATURAL IMAGES

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ABSTRACT

In this article, we propose a statistical model of natural images in JPEG format. The image acquisition is composed of three principal stages. First, a RAW image is obtained from sensor of Digital Still Cameras (DSC). Then, the RAW image is subject to some post-acquisition processes such as demosaicking, white-balancing and γ-correction to improve its visual quality. Finally, the processed images goes through the JPEG compression process. For each step of image processing chain, a statistical study of pixels' properties is performed to finally obtain a model of Discrete Cosine Transform (DCT) coefficients distribution.

Index Terms— Statistical model, Noise and system model, Natural image, JPEG format, DCT Coefficients.

1. INTRODUCTION

In the past decade, applications of computer vision for surveillance, security and safety have rapidly increased. The used images for these applications are usually compressed in JPEG format. The compression technology is a lossy process which help to reduce image size and can also strongly affect on the decision-making process. This explains why the study of image acquisition and processing chain from RAW format to JPEG format is of crucial importance.

Before display, a natural image acquired by a DSC undergoes some post-acquisition processes [1]. At this step, the initial RAW image which is obtained from Charge-Coupled Device sensor is not a full color image yet. The RAW image then goes through post-acquisition processes mainly composed of demosaicking, white balancing and γ-correction to obtain a TIFF image and improve its visual quality. Finally, the TIFF image goes through the JPEG compression process to reduce its size.

The main contribution of this paper is to provide a global distribution of DCT coefficients before quantization with a rigorously theoretical justification. The organization of the paper is as follows. A state of the art on statistical model of JPEG images is described in section 2. A statistical model of RAW image is explained in section 3 and extended in section 4 to the TIFF image. In section 5, the global model of DCT coefficients of JPEG image before the quantification is proposed. The section 6 presents some numerical results and the section 7 concludes the paper.

This work is partially supported by French National Agency (ANR) through ANR-CSOSG Program (Project ANR-007-SECU-004)

2. STATE OF THE ART

Over the past decades, there have been various studies on the distributions of the DCT coefficients for images. However, they have concentrated only on fitting the empirical data with a variety of well-known statistical distributions. Recently, Chang [2] has shown that the generalized Gamma model is more appropriate than the Laplacian model or the Generalized Gaussian model [3] to characterize the DCT coefficients. Particularly, Saad and others [3] have used the parameters of the Generalized Gaussian model to predict image quality scores. However due to the lack of mathematical analysis, a theoretical explanation of these various models of the DCT coefficients distribution is still missing. Lam and Goodman [4] showed that a doubly stochastic model of the images provides us an insight into the distribution. This model combines a Gaussian model for the DCT coefficients when the block variance is fixed and a model for block variance. However, it remains necessary to extend their work by studying the block variance to provide a universal model of DCT coefficients. This model can be obtained from the study of the whole processing chain of image acquisition.

3. MODEL OF RAW IMAGE

A. Foi et al [5] have studied and proposed an accurate model of a single RAW image. This RAW model is based on a Poissonian distribution modeling the photons sensing and a Gaussian distribution for the stationary disturbances in the output data. The normal approximation to the Poisson distribution gives a simplified model of the RAW pixels :

$$Z^c(m,n) \sim \mathcal{N}(\theta^c(m,n), S^c(m,n)) \tag{1}$$

where $Z^c(m,n)$ denotes the value of pixel at the location (m,n) for the color channel c (R, G, and B), θ^c and S^c are

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respectively its mean and variance. It can be noted that the mean and variance of the RAW pixel admit the following heteroscedastic relation:

$$S^c(m,n) = a \cdot \theta^c(m,n) + b \tag{2}$$

4. MODEL OF TIFF IMAGE

4.1. Normality and correlation in demosaicking and white balancing

In the scope of this paper, we study only the bilinear interpolation for the demosaicking algorithm and the Gray World for the white balancing algorithm.

For each color channel, the bilinear interpolation [1] calculates the missing value as a weighted mean of its neighbors which can be written as a linear filtering $Z^c_{DM} = h^c \circledast Z^c$ where h^c is the linear filter for color channel c and $\circledast$ denotes the 2-D convolution. Due to the linear property of the Gaussian distribution, the distribution of the pixel after the bilinear interpolation can be expressed as:

$$Z^c_{DM}(m,n) \sim \mathcal{N}(\theta^c_{DM}(m,n), S^c_{DM}(m,n)) \tag{3}$$

where $\theta^c_{DM} = h^c \circledast \theta^c$ and $S^c_{DM} = (h^c)^2 \circledast S^c$. As the missing values are interpolated from these neighbors, a pixel within a channel is correlated to its neighbors.

The white balancing process is implemented after the demosaicking process to correct illuminant light color modification. The Gray World algorithm [1] assumes that the average intensities of the red, green and blue channels should be equal. Therefore, the red and blue channel gains (the green channel is usually left unchanged) are calculated as:

$$C_{gain} = \mu^g_{DM} / \mu^c_{DM} \tag{4}$$

where μ^c_{DM} is the average intensity value for the color channel c. So the value of the red and the blue channel are modified by: $Z^c_{WB}(m,n) = C_{gain} \cdot Z^c_{DM}(m,n)$ For most of digital images, the number of pixels is sufficiently large to approximate $\mu^c_{DM} = \frac{1}{MN}\sum_{m=1}^{M}\sum_{n=1}^{N} Z^c_{DM}(m,n)$ by $\overline{\theta}^c_{DM} = \frac{1}{MN}\sum_{m=1}^{M}\sum_{n=1}^{N} \theta^c_{DM}(m,n)$. Using this approximation, the distribution of the pixel after the white balancing process can be expressed as:

$$Z^c_{WB}(m,n) \sim \mathcal{N}(\theta^c_{WB}(m,n), S^c_{WB}(m,n)) \tag{5}$$

where $\theta^c_{WB} = C_{gain} \cdot \theta^c_{DM}$ and $S^c_{WB} = C^2_{gain} \cdot S^c_{DM}$. Let notice that after the white balancing process, pixels distribution remains Gaussian with a correlation between neighbors due to demosaicking.

4.2. Non-normality in γ-correction

Finally, the γ-correction process necessary for contrast display purpose is given as [1]:

$$Z^c_{GM}(m,n) = |Z^c_{WB}(m,n)|^{\frac{1}{\gamma}} \tag{6}$$

This process involve a non-linear input-output mapping after which pixels are not Gaussian anymore. One can note that the pixel Z^c_{WB} is distributed as a Gaussian random variable, hence, $|Z^c_{WB}|$ has a folded normal distribution. By applying the change of variables theorem, the distribution of the pixel Z^c_{GM} after γ-correction can finally be written:

$$\begin{aligned} f(Z^c_{GM}) = \frac{\gamma (Z^c_{GM})^{\gamma-1}}{\sqrt{2\pi S^c_{WB}}} \Bigg[&\exp\Big(- \frac{((Z^c_{GM})^{\gamma} - \theta^c_{WB})^2}{2S^c_{WB}} \Big) \\ &+ \exp\Big(- \frac{((Z^c_{GM})^{\gamma} + \theta^c_{WB})^2}{2S^c_{WB}} \Big) \Bigg] \end{aligned} \tag{7}$$

where in an abuse of notation for the sake of clarity, $f(X)$ denotes the probability density function (pdf) of a random variable X.

5. DOUBLY STOCHASTIC MODEL OF DCT COEFFICIENTS

In the JPEG compression, the image should first be converted from RGB into a different color space called YCbCr. This transformation is linear , thus the statistical distribution of Y, Cb and Cr does not differ fundamentally from (7) and can easily be calculated at the cost of much complicated notations. For the sake of clarity, the distribution (7) is thus considered. Then, each channel (Y, Cb, Cr) must be divided into 8×8 blocks and each block is converted to a frequency domain representation using the DCT operation [4]:

$$\begin{aligned} I_{ij} = \frac{1}{4} T_i T_j \sum_{m=0}^{7}\sum_{n=0}^{7} Z^t(m,n) \\ \cdot \cos\left(\frac{(2m+1)i\pi}{16}\right) \cos\left(\frac{(2n+1)j\pi}{16}\right) \end{aligned} \tag{8}$$

where Z^t denotes the pixel for the color channel t (Y, Cb, Cr) and

$$T_k = \begin{cases} \frac{1}{\sqrt{2}} & \text{for } k = 0 \\ 1 & \text{for } k > 0 \end{cases}$$

Assuming Z^t pixels are identically distributed within a block, Lam and Goodman [4] confirmed that the model of DCT coefficients should be performed as a doubly stochastic model due to the variability of block variance. Using conditional probability, the global model of DCT coefficients can be expressed as:

$$f(I_{ij}) = \int_0^{\infty} f(I_{ij}|\sigma^2) f(\sigma^2) d(\sigma^2) \tag{9}$$

where σ^2 denotes the block variance.

Given a constant block variance, the DCT coefficients can be approximatively normally distributed by using the central limit theorem version for correlated random variables detailed in [6], which provides a result on asymptotic distribution of a sum of correlated variables. In our case, the variables are spatially correlated, which is a special case of the correlation studied in [6]. As the three first moments of each element are obviously finite, the distribution can be expressed as :

$$f(I_{ij}|\sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{I_{ij}^2}{2\sigma^2}} \tag{10}$$

Equations (9) and (10) highlight the crucial importance to analyze the distribution of the block variance $f(\sigma^2)$ in order to calculate the global distribution of DCT coefficients.

5.1. Block variance : Gamma model

The variance of block k can be expressed as :

$$\sigma_k^2 = \frac{1}{63} \sum_{m=0}^{7} \sum_{n=0}^{7} \left(Z_k^t(m,n) - \overline{Z_k^t} \right)^2 \tag{11}$$

where $\overline{Z_k^t}$ is the mean of pixels within block k, which is also a random variable. We have :

$$Z_k^t(m,n) - \overline{Z_k^t} = \frac{1}{64} \sum_{u=0}^{7} \sum_{v=0}^{7} \left(Z_k^t(m,n) - Z_k^t(u,v) \right) \tag{12}$$

By using the version of the central limit theorem from [6], $Z_k^t(m,n) - \overline{Z_k^t}$ can be approximated as a zero-mean Gaussian random variable. The square of a Gaussian variable is distributed as Chi-squared variable with one degree of freedom. Furthermore, a Chi-squared variable weighted by a positive constant is distributed as Gamma random variable $\mathcal{G}(\alpha, \beta_i)$ with shape parameter $\alpha = 1/2$. As a result, the block variance σ_k^2 can be considered as a sum of correlated Gamma variables.

It follows from [7] that the sum of correlated Gamma variables, denoted X, is identically distributed as the sum of independent Gamma variables, denoted Y, with different parameters due to the similar form of their moment-generating function. After then, the complicated distribution of Y can be approximated to a Gamma distribution by using the moment matching method. The two first positive moments of Y are:

$$\mathbb{E}[Y] = \alpha \sum_{i=1}^{n} \lambda_i \qquad \mathrm{Var}[Y] = \alpha \sum_{i=1}^{n} \lambda_i^2 \tag{13}$$

where (α, λ_i) are the parameters of each element in the sum Y. On the other hand, the moments of the Gamma distribution with parameters α^* and β^* are defined by :

$$\mathbb{E}[Y] = \alpha^* \beta^* \qquad \mathrm{Var}[Y] = \alpha^* (\beta^*)^2 \tag{14}$$

Therefore, the approximating Gamma distribution can be matched as

$$\alpha^* = \alpha \frac{(\sum_{i=1}^{n} \lambda_i)^2}{\sum_{i=1}^{n} \lambda_i^2} = \frac{(\sum_{i=1}^{n} \lambda_i)^2}{2\sum_{i=1}^{n} \lambda_i^2} \qquad \beta^* = \frac{\sum_{i=1}^{n} \lambda_i^2}{\sum_{i=1}^{n} \lambda_i} \tag{15}$$

As a result, the block variance can be modeled by the Gamma distribution $\mathcal{G}(\alpha^*, \beta^*)$.

5.2. Global model of DCT coefficients

It follows from (9) that the global distribution of DCT coefficients can be expressed as :

$$\begin{aligned} f(I_{ij}) &= \int_0^\infty \left(\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{I_{ij}^2}{2\sigma^2}} \right) \left(\frac{(\sigma^2)^{\alpha^*-1}}{(\beta^*)^{\alpha^*}\Gamma(\alpha^*)} e^{-\frac{\sigma^2}{\beta^*}} \right) d(\sigma^2) \\ &= \frac{1}{\sqrt{2\pi}(\beta^*)^{\alpha^*}\Gamma(\alpha^*)} \int_0^\infty e^{-\left(\frac{\sigma^2}{\beta^*} + \frac{I_{ij}^2}{2\sigma^2}\right)} (\sigma^2)^{\alpha^*-\frac{3}{2}} d(\sigma^2) \end{aligned} \tag{16}$$

From [8], the modified Bessel function is given by

$$K_\nu(xz) = \frac{z^\nu}{2} \int_0^\infty e^{-\frac{x}{2}\left(t + \frac{z^2}{t}\right)} t^{-\nu-1} dt \tag{17}$$

By replacing $x = \frac{2}{\beta^*}$, $z = |I_{ij}|\sqrt{\frac{\beta^*}{2}}$, $\nu = \frac{1}{2} - \alpha^*$, we obtain

$$\begin{aligned} K_{\frac{1}{2}-\alpha^*}\left(|I_{ij}| \sqrt{\frac{2}{\beta^*}} \right) &= \frac{1}{2} \left(|I_{ij}| \sqrt{\frac{\beta^*}{2}} \right)^{\frac{1}{2}-\alpha^*} \\ &\quad \cdot \int_0^\infty e^{-\left(\frac{\sigma^2}{\beta^*} + \frac{I_{ij}^2}{2\sigma^2}\right)} (\sigma^2)^{\alpha^*-\frac{3}{2}} d(\sigma^2) \end{aligned} \tag{18}$$

Denoting that $K_\nu(x) = K_{-\nu}(x)$, it follows from (16) and (18) the following distribution of DCT coefficients:

$$f(I_{ij}) = \frac{\sqrt{2}\left(|I_{ij}|\sqrt{\frac{\beta^*}{2}}\right)^{\alpha^*-\frac{1}{2}}}{\sqrt{\pi}(\beta^*)^{\alpha^*}\Gamma(\alpha^*)} K_{\alpha^*-\frac{1}{2}}\left(|I_{ij}|\sqrt{\frac{2}{\beta^*}}\right) \tag{19}$$

It can be noted that the Gaussian model and the Laplacian model are two special cases of this one. Indeed, when $\alpha^* = 1$, the Gamma distribution becomes the exponential distribution. Lam and Goodman [4] demonstrated how to obtain the Laplacian model from the exponential distribution of the block variance. When $\alpha^* \to \infty$, the block variance tends to zero, consequently the DCT coefficients tend to be distributed as Gaussian.

6. NUMERICAL RESULTS

We performed experiments on the Dresden Image Database [9] to verify the models presented above. The original RAW images were first converted using Dcraw (with parameters -D -4 -T -j -v) and then undergoes the whole processing described above. Figure 1 shows each stage of the image acquisition chain of a single image. It also shows that the model fit to the data is nearly perfect. In addition, we showed that the Laplacian model, which remains a popular choice in the literature, is not really adequate to model the DCT coefficients distribution. However, the Figure 1c doesn't show

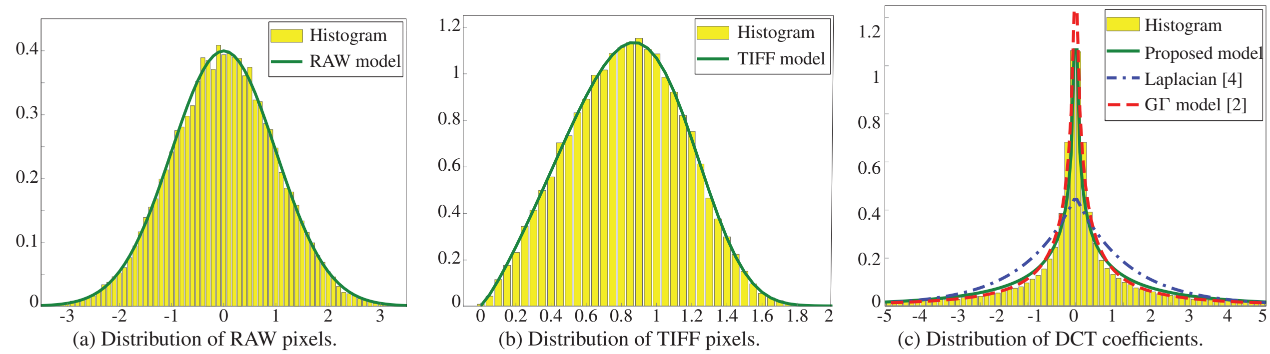

Fig. 1: Example of proposed model applied for natural image content estimation and noise whitening.

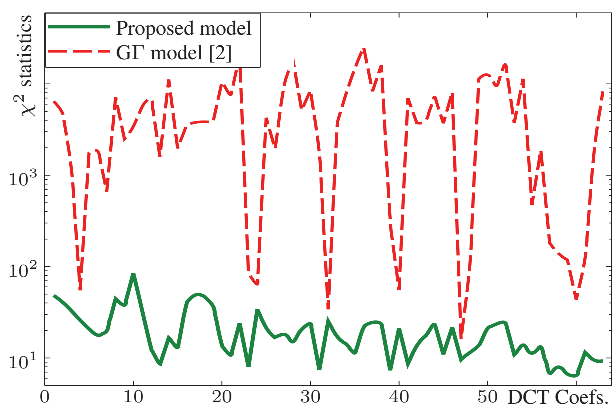

Fig. 2: χ^2 GOF test results of GΓ and proposed model.

the difference between the proposed model and the Generalized Gamma (GΓ) model. Consequently, we conducted a χ^2 goodness-of-fit (GOF) test to bring them into comparison. The parameters of the proposed model and the GΓ model were estimated using the Maximum Likelihood Estimation (MLE) approach. The experiments are conducted on all the images captured by *Nikon D200* of the Dresden Image Database. The Figure 2 shows the obtained mean result of χ^2 statistics for each DCT coefficient. The results show that the proposed model is more desirable than the GΓ model.

7. CONCLUSION

This paper provides a comprehensive insight about the image processing chain, from RAW format to JPEG format. We have proposed a global model of DCT coefficients and given an analytic justification of this model. We have shown that the proposed model of DCT coefficients is more appropriate than other model, in particular the GΓ model. The Laplacian distribution, which remains a popular choice in the literature, is also a special case of the proposed model.

Further research could look into the distribution of the quantized DCT coefficients so that a global model of the JPEG image can be established. This could give us a better insight into JPEG compression algorithm. This global model could b applied in various domains of image processing in the future.

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